

Module 2 Reflection Journal

Planning for Access, Agency, and Justice in Invisible Invaders

EDU 486 — Module 2

From Activities to an Access-to-Agency Pedagogy

A reading-to-practice reflection on how coherence, modeling, place, identity, access, technology, and justice shape Piter and Aastha's Invisible Invaders camp.

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Assignment Module 2 Reflection Journal
Course EDU 486 — Integrating Science and Technology
Camp Team Invisible Invaders — Piter and Aastha
Evidence Base Heather Table; Module 2 readings; July 14–21 class, planning, and youth-feedback records
Interpretive Lenses JuST Framework; Puzzle Plan and EQUITAS; UDL; PICRAT; justice-centered science
Submission Date July 22, 2026

Inclusion and rigor are the same design problem: both require us to make thinking visible, gather better evidence, and redesign barriers before explaining the learner as the problem.

This journal treats camp pedagogy as an interconnected learning system rather than a list of activities. The **Puzzle Plan** asks me to gather multiple forms of evidence, examine relationships among learners, tasks, tools, and environments, and revise instruction as new evidence appears. The **EQUITAS Access-to-Agency lens** asks whether those conditions create equitable, culturally responsive, identity-safe, and accountable routes for neurodivergent, chronically ill, disabled, multilingual, culturally and linguistically diverse, and multiply marginalized youth to participate, demonstrate competence, and exercise agency.

Journal Focus

At the beginning of Module 2, I had many activities in mind: collect sand, compare samples, inspect suspected particles, build a model, discuss injustice, and create art for advocacy. What I did not yet have was a clear pedagogy connecting them. That gap felt familiar because I have seen lessons look complete on paper while a learner's way into them remains invisible. Reviewing my Heather Table and the Module 2 readings shifted my attention from whether our schedule looked engaging to whether youth could see why each investigation mattered, bring their knowledge into the science, revise ideas with evidence, and choose how to act.

My central principle is that inclusion and rigor are the same design problem. As a Latino, bilingual, neurodivergent, and chronically ill educator, I know how quickly fatigue, processing time, uneven performance, or a communication difference can be mistaken for a lack of effort or ability. That is why I need to make the intellectual work visible, gather better evidence of what youth are thinking, and revise the environment before I explain the learner as the problem. My Puzzle Plan and EQUITAS Access-to-Agency work gave me that commitment; Module 2 gave me more concrete ways to practice it.

For our Invisible Invaders camp, this means organizing the week around the anchoring phenomenon, *"Humans aren't trash cans, but we have plastics in our bodies,"* and the driving question, *"How does plastic break into tiny pieces, move through the environment and into living bodies, and how can we stop it fairly?"* The phenomenon invites investigation, but the way we facilitate that investigation will determine whether youth experience science as something they help build or as another set of conclusions delivered to them.

Principles at a Glance

Principle	Reading roots	Visible camp decision
Coherence belongs to learners	Reiser et al.; Cherbow et al.; Wiggins and McTighe	Keep the driving question, complete sequence, and Four Gotta-Haves visible; give one direction at a time; preserve pause and re-entry.
Models are public and revisable	Windschitl and Thompson; Ippolito et al.; Smetana et al.; Wisbrock et al.	Move from controlled jars and field evidence to a source-to-body model; label claims supported, possible, or unknown; revise instead of replacing youth thinking.
Place and community knowledge are science	Seiki et al.; Starr et al.; Nesmith et al.	Treat Charlotte Beach and youth-known places as evidence; examine infrastructure and power; invite community connections without requiring disclosure.
Identity and joy belong inside rigor	Mercier and Carlone; Muhammad	Recognize investigating, designing, connecting, caring, revising, and advocating; notice quiet contributions; provide movement and low-energy routes.
Access must be designed into thinking	Liljedahl; Kimmons et al.; CAST	Preserve demanding sensemaking without requiring one posture, pace, group, or communication mode; use technology only for access, evidence, or agency.
Justice links mechanism, power, and action	Smetana et al.; Wisbrock et al.; Nesmith et al.; Muhammad	Connect plastic pathways to production, infrastructure, policy, and unequal burdens; support collective action without blame or unsupported health claims.

Principle 1: Coherence Must Be Visible to Learners

Reiser, Novak, McGill, and Penuel (2021) argue that a storyline should be coherent from the students' perspective: youth questions and emerging explanations should create a felt need for the next investigation. Cherbow, McNeill, Lowell, and Grymonpre (2026) press this point further by showing that a well-designed curriculum cannot guarantee a coherent experience. Teachers still have to build the relationship among students, curricular resources, and the evolving storyline. Backward design helps me identify the final understanding and acceptable evidence before choosing activities (Wiggins & McTighe, 2005),¹ but a backward plan remains only my plan unless youth can see and influence the path.

For Invisible Invaders, coherence is now something youth can see, not something hidden in our planning document. In my elementary technology placement, I saw students understand a coding idea while a long verbal direction or unfamiliar login path hid that understanding. Modeling the tool path and showing a first/next/last sequence made the route visible without making the idea easier. For camp, we will keep the driving question, the day's purpose, the activity sequence, and our Four Gotta-Haves visible. Each connecting question must help youth explain one part of the phenomenon.

This principle changed after meeting the Freedom Scholars and staff on July 21. Youth asked for short explanations, one direction at a time, and a chance to begin doing quickly. That feedback did not ask us to simplify the science. It asked us to remove the executive-function burden of holding an invisible sequence in working memory. A visual agenda, a completed starting example, and brief demonstrations make the causal problem easier to enter while leaving the thinking demanding. When a learner joins late, takes a break, or needs to re-enter after pain, sensory overload, or attention loss, the storyline remains available without requiring a public explanation.

Principle 2: Models Are Public, Revisable Thinking

The modeling toolkit helped me move beyond treating a model as a polished poster made at the end of a unit. Windschitl and Thompson (2013) frame models as public representations that make both observable events and unobservable mechanisms available for critique and revision. Science and literacy also strengthen one another when learners observe, investigate, talk, read, write, and communicate for an authentic purpose rather than completing disconnected language exercises (Ippolito, Condie, Blanchette, & Cervoni, 2018). Local litter inquiries show how firsthand evidence can lead toward explanation and public communication (Smetana, Wisbrock, Zaker, & Hoellein, 2019; Wisbrock et al., 2020).

Our camp model must preserve change, uncertainty, and evidence limits. On Field Friday, youth will compare controlled jars and examine traceable Charlotte Beach sand samples; on Monday, they will build a first source-to-body model; and on Thursday, they will revise a shared model with evidence gathered during the week. During our July 14 class, I argued that disagreement in a consensus model should push us toward deeper questions and reveal whose voice is being heard or excluded, not be erased for the sake of agreement. Friday's KEEP, QUESTION, CHANGE routine protects that kind of evidence-based revision.

This also requires honest claim language. A visually suspected particle is not chemically confirmed plastic. Finding microplastics in a body or food source establishes exposure, but it does not by itself prove a particular route or health effect. Youth can label claims as *supported*, *possible*, or *still unknown*. Those labels are not signs that the investigation failed. They are part of learning how science protects people from false certainty and how responsible advocacy can remain strong without exaggeration.

I do not want fast speech or comfort with public presentation to become our evidence of who understands. A learner may contribute by speaking, pointing, drawing arrows, writing a label, dictating to a partner, taking an

¹The Module 2 schedule labels this reading as 2021; the assigned course PDF is the 2005 expanded second edition cited here.

evidence photograph, or asking to pass and return. These are not easier versions of modeling. Each route still asks the learner to connect evidence, mechanism, and claim; only the form of the representation changes.

Principle 3: Place, Community, and Family Knowledge Are Science

Seiki, Domínguez, and Asato (2021) describe border crossing between home and school by positioning familial knowledge alongside formal science rather than asking learners to leave one world behind. Starr et al. (2026) similarly argue that outdoor learning is not just science moved to a different location. Place changes relationships among youth, families, educators, histories, and socioecological systems. The transdisciplinary river course described by Nesmith et al. (2021) shows why environmental problems cannot be understood through science content alone; policy, economics, ethics, art, history, and community knowledge shape both the problem and possible action.

Charlotte Beach and the places youth already know are not scenery for our lesson; they are part of the evidence and the question of responsibility. As a bilingual learner, I know an idea can sound less complete when someone is translating language, culture, and academic expectations at the same time. I do not want academic English to become a substitute for scientific credibility. Youth may connect the model to a neighborhood, bus route, laundry practice, food system, family story, or familiar object through speech, home language, drawing, or partner talk. We will ask who makes decisions, who benefits, who carries burdens, and whose knowledge is recognized without using private addresses or ZIP codes as shortcuts for identity or risk.

As someone who lives with chronic illness, I know a question about bodies can become personal very quickly. I need to treat connection as an invitation, not a demand for disclosure. No learner should have to share a diagnosis, body history, or family experience to prove that plastics pollution matters. Youth can work through an invented scenario, public place, provided image, or community-scale example. That choice protects identity safety while keeping community knowledge available as an intellectual resource.

Principle 4: Identity, Recognition, and Joy Belong Inside Rigor

Mercier and Carlone (2021) show how identi-beads and identi-badges can help youth recognize multiple forms of STEM participation, including investigating, inventing, designing, tinkering, conservation, and care for others. The important move is not the bead itself. It is making a wider range of contributions visible to oneself and to the group. Muhammad (2023) brings identity, skills, intellect, criticality, and joy into one curricular frame, reminding me that justice-centered learning cannot stop after naming harm.

I want every Freedom Scholar to leave with evidence that their way of contributing mattered to the science community we built. I know what it feels like to be noticed most clearly at the point of difficulty while the reasoning, persistence, and work behind it remain invisible. We will therefore notice careful observation, revision, connection-making, advocacy, creativity, care, and questions that redirect the group, including quiet and behind-the-scenes contributions. A bead, badge, or acknowledgment should help youth name a contribution they value in themselves, not rank them or decide which kind of STEM person they are allowed to be.

Joy, in this camp, is not a prize added after rigorous work. It appears when youth experience curiosity, movement, humor, friendly challenge, beauty in advocacy art, meaningful relationships, and the power to imagine a more just future. The students' feedback made this practical: include movement and social connection, but also a seated or lower-energy path; show an example without requiring imitation; and give early finishers a meaningful extension instead of more repetitive work. Joy becomes compatible with chronic illness and neurodivergence only when participation does not require performing endless energy.

Principle 5: Access Must Be Built Into the Thinking Environment

Liljedahl (2020) challenged me to examine classroom routines that produce compliance without much thinking. I value the emphasis on rich tasks, visible collaborative work, carefully chosen hints and extensions, and teachers answering questions in ways that return agency to learners. PICRAT adds a related technology question: What relationship does the learner have with the tool, and does the tool replace, amplify, or transform the practice (Kimmons, Graham, & West, 2020)? UDL reinforces planning for multiple ways to engage, perceive information, and express understanding from the beginning (CAST, 2024).

However, no single posture, communication mode, group arrangement, or technology can stand in for thinking. A vertical whiteboard may make ideas public for one learner and create pain, fatigue, motor, sensory, or social barriers for another. Random grouping may widen collaboration, but it can also remove a stabilizing relationship. Verbal questioning may reveal reasoning, or it may measure processing speed and speech access. My responsibility is to preserve the goal of visible, collaborative thinking while offering standing, seated, tabletop, visual, verbal, partner, photo, and low-energy routes.

This is where my prior work presses against a mechanical use of any framework. In my placement, one learner stayed connected to a building activity while keeping a familiar laptop nearby. Treating the device as support rather than distraction reminded me that regulation can make thinking possible. During camp, Aastha and I will check in proactively, give one direction at a time, keep the sequence visible, make public reading and presenting optional, and let youth pass and re-enter. These supports do not lower the evidence expectation; they help us gather better evidence of what youth understand.

Technology will remain optional and purposeful. A phone camera can preserve a transient observation, zoom in on a suspected fiber, or let a learner contribute without copying a large amount of text. A shared slide or projected model can make revisions visible. A timer can support transitions. These tools aren't automatically just because they're digital. I'll ask whether a tool increases access to the phenomenon, strengthens evidence, or expands youth agency. If paper, a magnifier, or conversation does a better job, technology shouldn't displace it.

Principle 6: Justice Must Move From Evidence to Collective Action

The litter investigations connect local evidence with communication and action, while the transdisciplinary and joy-centered readings keep the analysis from becoming a narrow cleanup lesson (Muhammad, 2023; Nesmith et al., 2021; Smetana et al., 2019; Wisbrock et al., 2020). If we ask only, "What can an individual camper recycle?" we hide the roles of product design, production, labor, wastewater systems, policy, corporate decisions, and unequal exposure. If we describe harm without opening a path toward change, we can leave youth with fear or helplessness.

Justice-centered science must help youth connect mechanism, power, and an action that does not blame the people carrying the greatest burden. I have experienced institutions place responsibility on an individual while leaving an inaccessible system unquestioned, so I do not want camp to stop at telling youth to recycle better. Our model will identify where plastic is produced, released, transported, encountered by living systems, and governed by human choices. Youth advocacy can address product design, reuse and repair, monitoring, public communication, research, or policy, while art lets them argue for the future they want.

The action still has to match the evidence. We should not use a body-based anchor to frighten youth or make unsupported health claims. We should not pretend that cleaning one sample solves a system-level problem. Instead, youth can say what the evidence supports, what remains uncertain, who needs better information, and what change would reduce a pathway or distribute responsibility more fairly. That is a stronger form of advocacy because it treats the audience and the young scientist with respect.

My Camp Commitments

These principles leave me with a practical set of commitments for Invisible Invaders:

- 1. Plan backward, then revise forward.** I will keep the final explanation and evidence goals clear while changing the route in response to youth questions, observations, access needs, and feedback.
- 2. Keep the storyline visible.** Every investigation will connect to the driving question and a named Gotta-Have, with the purpose and sequence available for learners who need to pause or re-enter.
- 3. Protect scientific uncertainty.** Models and public products will distinguish observation from inference, suspected from confirmed particles, and exposure evidence from claims about route or harm.
- 4. Design equivalent routes into rigor.** Movement and seated options, speaking and non-speaking options, individual thinking and supported collaboration will all lead to evidence-based explanation and action.
- 5. Track whose thinking becomes public.** I will notice whose questions, model relationships, and proposed actions enter the shared record, especially when a learner contributes quietly or outside the expected form.
- 6. Use technology for access, evidence, or agency.** Novelty is not a learning goal. A tool earns its place when it makes meaningful scientific participation possible or more powerful.
- 7. Debrief like a learning scientist.** At the end of each day I will ask what youth evidence changed our plan, which barrier hid someone's thinking, and what Aastha and I need to redesign before the next session.

Closing Reflection

Module 2 did not teach me that access matters; my life and teaching had already made that impossible to ignore. It gave me a more disciplined way to act on that knowledge. Coherence asks whether youth can understand and shape the journey; modeling, whether their ideas become public and revisable; place and familial curriculum, whose knowledge enters science; identity and joy, who is recognized as belonging; UDL, PICRAT, and my Puzzle Plan, whether the environment reveals thinking or filters learners through one preferred bodymind; and justice, whether evidence opens power-aware, collective possibility.

The teacher-as-learning-scientist stance begins by studying the learning environment before explaining the learner as the problem. In Invisible Invaders, I want that stance to be visible in small choices: one direction at a time, a model that can change, an evidence label that protects uncertainty, an optional public presentation, a proactive check-in, and more than one credible way to act. I want campers to remember that their questions changed our model, their way of participating counted, and uncertainty did not make their evidence worthless. That is how I will know whether our rigorous science became accessible enough for youth to claim as their own.

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